Econ 812 Recitation 6 Notes

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1 Some notes from class

For a predetermined variable k_t and a jump variable y_t we have

$$x_t = \begin{bmatrix} k_t \\ y_t \end{bmatrix}, x_t^* = \begin{bmatrix} s_t \\ u_t \end{bmatrix}$$

Two key equations are:

$$\begin{bmatrix} s_t \\ u_t \end{bmatrix} = x^* = P^{-1}x_t = P^{-1}\begin{bmatrix} k_t \\ y_t \end{bmatrix} = \begin{bmatrix} R_{sk} & R_{sy} \\ R_{uk} & R_{uy} \end{bmatrix}\begin{bmatrix} k_t \\ y_t \end{bmatrix}$$

and

$$\begin{bmatrix} s_{t+1} \\ u_{t+1} \end{bmatrix} = \begin{bmatrix} J_s & 0 \\ 0 & J_u \end{bmatrix}\begin{bmatrix} s_t \\ u_t \end{bmatrix} + P^{-1}A^{-1}Cz_t$$

2 Fall 22 Midterm Question 3

Solving linear difference equations: Suppose we have the following system

$$\begin{bmatrix} \hat{y}_{t+1} \\ \hat{p}_{t+1} \end{bmatrix} = \begin{bmatrix} 6 & -4 \\ -1 & 6 \end{bmatrix}\begin{bmatrix} \hat{y}_t \\ \hat{p}_t \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} \hat{z}_{t+1} \\ \hat{z}_t \end{bmatrix} \tag{1}$$

The eigenvalues of W are 4 and 8. The corresponding matrix of eigenvectors is

$$P = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \text{ and } P^{-1} = \begin{bmatrix} 1/4 & 1/2 \\ -1/4 & 1/2 \end{bmatrix}$$

Suppose that y_t and p_t are endogenous jump variables.

2.1 Part A

Does the system have a unique bounded solution for any bounded $\{\hat{z}_t\}$? Explain.
Yes, from the theorem in class...

- $n_u = 2 = n_y + n_p$

- $R_{uy} = P^{-1}$ which is invertible
- $|\lambda_i| \neq 1 \forall i$

So there is a unique bounded solution.

2.2 Part B

Suppose $\hat{z}_t = (1/4)^t$. Solve for the unique bounded solution $\hat{y}_t$ and $\hat{p}_t$ to the system.

$$P \begin{bmatrix} u_{t,y} \\ u_{t,p} \end{bmatrix} = \begin{bmatrix} y_t \\ p_t \end{bmatrix}$$

$$\begin{bmatrix} u_{t+1,y} \\ u_{t+1,p} \end{bmatrix} = J \begin{bmatrix} u_{t,y} \\ u_{t,p} \end{bmatrix} + P^{-1}A^{-1}C \begin{bmatrix} \hat{z}_{t+1} \\ \hat{z}_t \end{bmatrix}$$

We have that $P^{-1}A^{-1}C = \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & -1/2 \end{bmatrix}$ So we can write:

$$u_{t+1,y} = 4u_{t,y} + 1/2\hat{z}_{t+1} - 1/2\hat{z}_t$$

$$u_{t+1,p} = 8u_{t,p} + 1/2\hat{z}_{t+1} - 1/2\hat{z}_t$$

Then we also have that $\hat{z}_{t+1} = 1/4\hat{z}_t$. Substitute to find that:

$$u_{t,y} = 1/4u_{t+1,y} + 3/32\hat{z}_t$$

$$u_{t,y} = \frac{3}{32}\sum_{j=0}^{\infty}(1/4)^j\hat{z}_{t+j} = \frac{3}{32}\sum_{j=0}^{\infty}(1/4)^j(1/4)^j\hat{z}_t = \frac{3}{32}\frac{1}{1-1/16}\hat{z}_t = 1/10\hat{z}_t$$

Repeat a similar process for p, and plug in to solve for $\hat{y}_t$ and $\hat{p}_t$

3 Modified Aug 23 Comp

Suppose we have log-linearized our model to get the following system:

$$\begin{bmatrix} \hat{K}_{t+1} \\ \hat{C}_{t+1} \end{bmatrix} = \begin{bmatrix} 1.1 & -0.3 \\ -0.1 & 1 \end{bmatrix} \begin{bmatrix} \hat{K}_t \\ \hat{C}_t \end{bmatrix} + \Gamma\hat{G}_t, \text{ with } \Gamma = \begin{bmatrix} -0.0661 \\ 0.0034 \end{bmatrix} \tag{2}$$

Let

$$J = \begin{bmatrix} 0.9 & 0 \\ 0 & 1.2 \end{bmatrix}, P = \begin{bmatrix} 0.9 & 1 \\ 0.5 & -0.3 \end{bmatrix}, P^{-1} = \begin{bmatrix} 0.4 & 1.3 \\ 0.7 & -1.2 \end{bmatrix}, P^{-1}\Gamma = \begin{bmatrix} -0.0205 \\ -0.0503 \end{bmatrix}$$

and

$$\hat{G}_{t+1} = 0.9\hat{G}_t$$

3.1 Part A

Is there a unique bounded solution?

Yes there is since this is a neoclassical model with government spending shocks, $n_u = n_y = 1$, $|\lambda_i| \neq 1 \forall i$ and $R_{uy} = -1.2$ which is invertible.

3.2 Part B

Solve for equilibrium $\hat{C}_t = \mu_{KC}\hat{K}_t + \mu_{GC}\hat{G}_t$

$$\begin{bmatrix} s_{t+1} \\ u_{t+1} \end{bmatrix} = J \begin{bmatrix} s_t \\ u_t \end{bmatrix} + P^{-1}\Gamma\hat{G}_t$$

$$u_{t+1} = 1.2u_t - 0.0503\hat{G}_t$$

This solves to

$$u_t = 0.0503/1.2 \sum_{j=0}^{\infty} (\frac{1}{1.2})^j 0.9^j \hat{G}_t = \frac{0.0503}{1.2} \frac{1}{1-0.75}\hat{G}_t$$

and with

$$\begin{bmatrix} s_t \\ u_t \end{bmatrix} = P^{-1} \begin{bmatrix} K_t \\ C_t \end{bmatrix}$$

Have that

$$u_t = 0.7\hat{K}_t - 1.2\hat{C}_t$$

So combining to get

$$\frac{0.0503}{1.2} \frac{1}{1-0.75}\hat{G}_t = 0.7\hat{K}_t - 1.2\hat{C}_t$$

$$\hat{C}_t = -\frac{0.0503}{1.2^2} \frac{1}{1-0.75}\hat{G}_t + 0.7\hat{K}_t$$

4 Another example with identifying R_{uy}

This problem was from June 2017 Comp. Identify R_{uy}

$$R = P^{-1} = \begin{bmatrix} 1.6 & 1.2 & -1.8 & 3.5 \\ 2.6 & 0.3 & -0.1 & 4.7 \\ -0.16 & 2 & -1.3 & -2.3 \\ -0.6 & -0.6 & 0.5 & 1.7 \end{bmatrix}$$

$$J = \begin{bmatrix} 1.7 & 0 & 0 & 0 \\ 0 & 1.2 & 0 & 0 \\ 0 & 0 & 0.6 & 0 \\ 0 & 0 & 0 & 0.8 \end{bmatrix}$$

$$x_t = \begin{bmatrix} k_t \\ y_t \end{bmatrix} = \begin{bmatrix} k_{1t} \\ k_{2t} \\ y_{1t} \\ y_{2t} \end{bmatrix}$$

R_{uy} is the top right R_{uk} is the top left R_{sk} is the bottom left and R_{sy} is the bottom right.